

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                             |   | Marks available |     |     |       |       |      |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                             |   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8        | (a) | (i)  |  | as pH increases, citric acid content decreases and sugar content increases                                                                                                                                                                                                                                                                                  | ✓ |                 |     | 2   | 2     | 2     |      |
|          |     |      |  | as acidity decreases, ascorbic acid content decreases and water content decreases                                                                                                                                                                                                                                                                           |   |                 |     |     |       |       |      |
|          |     |      |  | tomatoes are a good source of vitamin C and citric acid                                                                                                                                                                                                                                                                                                     |   |                 |     |     |       |       |      |
|          |     |      |  | citrus fruits contain ascorbic acid and a natural preservative                                                                                                                                                                                                                                                                                              | ✓ |                 |     |     |       |       |      |
|          |     | (ii) |  | award (1) for <b>any</b> of following <ul style="list-style-type: none"><li>the values for ascorbic acid are bigger but the unit is 1000 times smaller</li><li>mg/100g is a smaller unit than %</li><li>citric acid is measured in % but ascorbic acid is measured in mg/100g</li><li>there is much more citric acid (than ascorbic acid) present</li></ul> |   |                 |     | 1   | 1     | 1     |      |
|          | (b) | (i)  |  | $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$<br>formulae (1)<br>state symbols (1)                                                                                                                                                                                                                              |   | 2               |     |     | 2     |       |      |

| Question |  |       |    | Marking details                                                                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|--|-------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |       |    |                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (ii)  | I  | $0.35 \times 0.021 = 0.00735$                                                                                                                                                                                                                                                                       |                 | 1   |     | 1     | 1     | 1    |
|          |  |       | II | $0.588 / 0.59$ (2)<br>if answer incorrect credit each correct step in method<br>$0.00735 \times 2 = 0.0147$ (1)<br>$\frac{0.0147}{0.025} = 0.588 / 0.59$ (1)<br>ecf possible from part (i)                                                                                                          |                 | 2   |     | 2     | 2     | 2    |
|          |  | (iii) |    | $1.2 \times 10^{22}$ (2)<br>accept $12 \times 10^{21} / 0.12 \times 10^{23}$<br>if answer incorrect credit each correct step in method but do not award any marks for multiplying mass 0.36 by $N_A$<br>$\frac{0.36}{18} = 0.020$ (1)<br>$0.020 \times 6.0 \times 10^{23} = 1.2 \times 10^{22}$ (1) |                 | 2   |     | 2     | 2     |      |
|          |  |       |    | Question 8 total                                                                                                                                                                                                                                                                                    | 2               | 5   | 3   | 10    | 8     | 3    |